QF600 Asset Pricing — Lecture 7 Reference Cheatsheet

Multi-Period Asset Pricing: wealth dynamics, the lifetime problem, the value function, the Bellman equation, backward induction, the multi-period stochastic discount factor

Built from Lecture 7 slides (Liu Peng, LKCSB) · October 2026 · with symbol-by-symbol equation breakdowns

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Source: "Lecture slides – Week 7 v2" (Liu Peng, Lee Kong Chian School of Business, 13 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. △**!** marks a slide ambiguity or error, with the corrected version.

How to read the equations: every displayed equation is followed by a shaded where box that names each symbol and component and says what it represents in that formula.

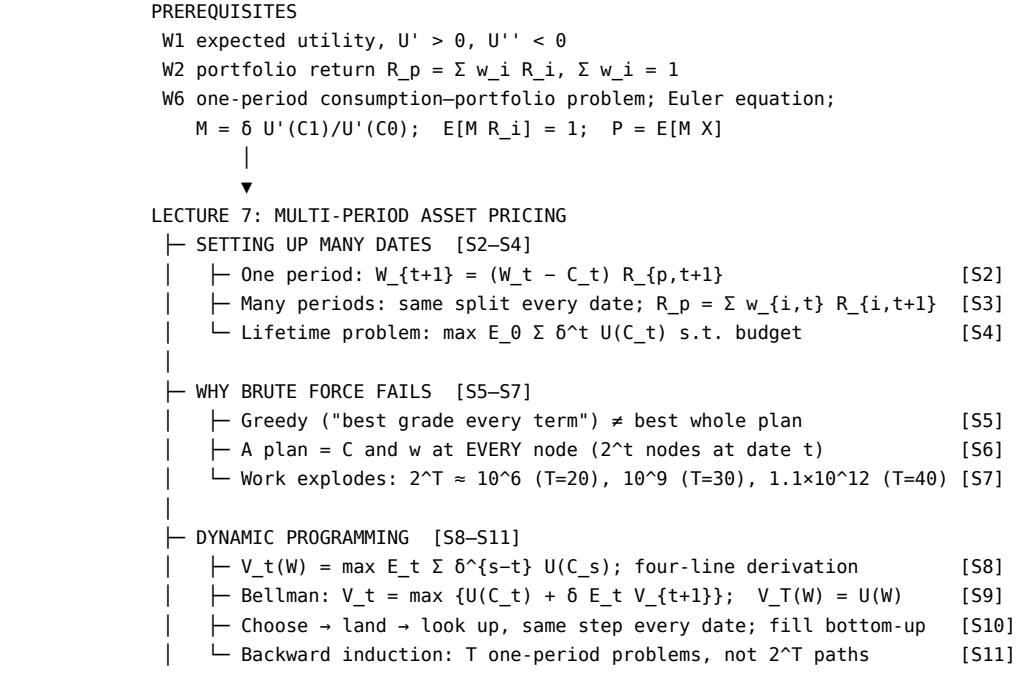
Exam context (Week 1 [S5]): closed book, one double-sided A4 sheet allowed; 40% MCQ and 60% written (concepts plus simple calculations), with **no coding**. From this week you need: the wealth-evolution equation; the lifetime problem and what a "plan" is; the four-line derivation of the Bellman equation; the terminal condition $V_T(W) = U(W)$; why backward induction replaces 2^T paths with T one-period problems; and the telescoping kernel $M_{t,t+i} = \delta^i U'(C_{t+i})/U'(C_t)$. The *[Beyond slides]* items (envelope condition, conditional Euler equation, bond prices, the log-utility closed form) are the standard next steps.

Notation used throughout (slide conventions)

Symbol	Meaning	Type	Notes
$t = 0, 1, \ldots, T$	dates; T is the last date (horizon)	time	$T+1$ dates, T periods
W_t	wealth at the start of date t	\$	W_0 given [S4]
C_t	consumption at date t (a choice)	\$	$C_t \geq 0$ [S4]
$W_t - C_t$	amount invested at t	\$	only this earns a return [S2]
$w_{i,t}$	weight of asset i chosen at t	fraction	$\sum_i w_{i,t} = 1$ [S4]
$R_{i,t+1}$	gross return on asset i from t to $t+1$	random	"what \$1 becomes"
$R_{p,t+1}$	portfolio gross return $\sum_i w_{i,t} R_{i,t+1}$	random	[S3]
$U(\cdot)$	period utility, same every date	function	$U' > 0$, $U'' < 0$ (W6)
δ	time-preference factor	number	δ^t: "later counts for less" [S4]
$\mathbb{E}_t[\cdot]$	expectation given information at t	operator	$\mathbb{E}_t[\mathbb{E}_{t+1}[\cdot]] = \mathbb{E}_t[\cdot]$ [S8]
$V_t(W)$	value function: best expected lifetime utility from t on, given wealth W	function	[S8]–[S9]
M_{t+1}	one-period SDF $\delta U'(C_{t+1})/U'(C_t)$	random	W6's $\tilde{M}$ with dates [S12]
$M_{t,t+i}$	i-period SDF (kernel)	random	$M_{t+1} \cdots M_{t+i}$ [S13]
$X_{t+i},\ P_t$	payoff at $t+i$, its price at t	\$	*[Beyond slides]*
$B_t^{(i)}$	price at t of a zero paying \$1 at $t+i$	\$	$\mathbb{E}_t[M_{t,t+i}]$ *[Beyond slides]*
*	evaluated at the optimum	–	e.g. C_t^*

Notation trap: Week 6 marked random variables with a tilde; this deck drops it, so anything dated after the conditioning date of $\mathbb{E}_t$ is random. Here W_t is *wealth*; in QF620 W_t is a Brownian motion. Nothing this week is continuous-time.

0. Topic map



```
│
├─ PRICING OVER MANY PERIODS  [S12–S13]
│   ├─ M_{t+1} M_{t+2} = δ² U'(C_{t+2})/U'(C_t)  (telescoping)        [S12]
│   └─ M_{t,t+i} = δ^i U'(C_{t+i})/U'(C_t): time pref × state adj.    [S13]
│
└─ CONSEQUENCES  [Beyond slides]
    ├─ Envelope V_t'(W) = U'(C_t*)  ⇒  E_t[M_{t+1} R_{i,t+1}] = 1 every date
    ├─ P_t = E_t[M_{t,t+i} X_{t+i}];  B_t^(i) = E_t[M_{t,t+i}]
    └─ Log utility: C_t = W_t (1−δ)/(1−δ^{T−t+1}); myopic portfolio
       │
       ▼
NEXT: dividend streams, the term structure, consumption-based tests (equity premium puzzle)
```

Dependency logic:

- The setup repeats Week 6's one-period budget at every date; the new feature is that today's wealth depends on all past choices and returns.
- You can't optimise one date at a time (greedy fails) and you can't enumerate every path (the tree explodes).
- The value function summarises "the best you can do from here" as one function of wealth, so each date becomes a Week 6 one-period problem with V_{t+1} in place of U.
- The pricing side is the same idea: one-period SDFs link consecutive dates, and chaining links prices any horizon.

1 Concept entries

1.1 One period: eat or invest [S2]

$$W_{t+1} = (W_t - C_t) \cdot R_{p,t+1}.$$

where

W_t	wealth at date t.
C_t	consumption at t: "eaten today. Gone — it never compounds."
$W_t - C_t$	what is left over and invested.
$R_{p,t+1}$	portfolio *gross* return from t to $t+1$ (a 6% gain is 1.06); chosen at t, realised at $t+1$.
W_{t+1}	wealth at $t+1$: only the invested part is multiplied by the return.

- Identical to W6's $\tilde{C}_1 = (W_0 - C_0)\tilde{R}_p$, except that date-$(t+1)$ wealth is split again instead of all eaten.
- **Traps:** the return multiplies $(W_t - C_t)$, not W_t; use gross, not net, returns; $R_{p,t+1}$ is random at t.

1.2 Many periods: wealth evolution [S3]

$$W_{t+1} = (W_t - C_t) \cdot R_{p,t+1}, \qquad R_{p,t+1} = \sum_i w_{i,t} R_{i,t+1}.$$

where

$w_{i,t}$	weight of asset i, chosen at date t (can change every period: rebalancing).
$R_{i,t+1}$	gross return of asset i over $[t, t+1]$.
$\sum_i w_{i,t} R_{i,t+1}$	the portfolio return is the weighted average of asset returns.

- "Each date repeats the same split: eat some, invest the rest, and start the next date with whatever the portfolio returned."
- *[Beyond slides]* With no consumption in between, $W_{t+i} = W_t R_{p,t+1} \cdots R_{p,t+i}$: multi-period gross returns *multiply*.
- W_t is the **state variable**: it carries everything about the past that matters for the future. That is what makes the recursion in §1.6 work.

1.3 The lifetime problem [S4]

$$\max_{C,w} \ \mathbb{E}_0\Big[\sum_{t=0}^{T} \delta^t U(C_t)\Big] \quad \text{s.t.} \quad W_{t+1} = (W_t - C_t)\sum_i w_{i,t} R_{i,t+1}, \quad \sum_i w_{i,t} = 1, \quad W_0 \text{ given}, \ C_t \geq 0, \ W_t \geq 0.$$

where

$\max_{C,w}$	"what you pick, at every date": a consumption and weight choice for every date and state.
$\mathbb{E}_0$	"averaged over every state", from the date-0 viewpoint.
δ^t	"later counts for less": utility at t is discounted t times.
$U(C_t)$	"value of eating at t".
$W_t \geq 0$	no dying in debt (finite-horizon no-Ponzi condition) *[Beyond slides]*.

- **Key message [S4]:** "One rule for every state, chosen once — not a number chosen each morning." The choices are *functions* of the history (a contingent plan), not single numbers.
- **Assumptions:** time-separable expected utility (W6 [S10]), same U and constant δ, no labour income, finite T, no bequest.
- $T = 1$ gives back Week 6 exactly.
- **Trap:** δ^t discounts *utility*, not dollars; the dollar discount is the SDF (§1.10).

1.4 Greedy is not optimal [S5]

- Picture: "Grade chasing" is top of the class early, then flattens and falls; "Whole plan" starts lower and ends far higher by age 40. "The grade you give up early buys the skill the first job pays for."
- Maximising each period's payoff separately is not the same as maximising the sum.
- *[Beyond slides]* In asset pricing: eating more today raises $U(C_t)$ but lowers W_{t+1} and so every later option. The right one-period problem includes $\delta\mathbb{E}_t[V_{t+1}]$, the value of what you leave yourself.

1.5 The plan lives on a tree; the work explodes [S6–S7]

- **Tree [S6]:** two outcomes per date gives 1 node at $t = 0$, 2 at $t = 1$, 4, 8, ..., 2^T at $t = T$. "A plan names C and w at every node — not just along the path that happens."
- **Explosion [S7]:**

Horizon T	2^T	Slide label
10	1,024	(unlabelled point)
20	1,048,576	"1 million nodes"
30	$\approx 1.07 \times 10^9$	"1 billion nodes"
40	$\approx 1.10 \times 10^{12}$	"1.1 trillion nodes"

- *[Beyond slides]* The full tree up to T has $2^{T+1} - 1$ nodes; with n outcomes per date, n^T.
- Exponential in T for brute force vs linear in T for backward induction (§1.8).

△! [S7] The curve is not drawn to scale: each +10 in T multiplies the count by $2^{10} \approx 1000$, so on a true scale the earlier points would sit on the axis. Read the labels, not the heights.

1.6 The value function and the Bellman equation [S8–S9]

Definition [S8], line 1:

$$V_t(W_t) = \max_{C,w} \mathbb{E}_t\Big[\sum_{s=t}^{T} \delta^{s-t} U(C_s)\Big].$$

where

$V_t(W_t)$	best expected discounted utility from t onwards, given wealth W_t.
$\max_{C,w}$	over the whole remaining plan $C_t, \ldots, C_T$ and weights.
s	running date, from today t to the end T.
δ^{s-t}	discounting restarts from today: date t gets $\delta^0 = 1$.

Derivation (exam-reproducible) [S8]:

1. Definition above.
2. **Split the sum.** Once C_t is chosen, today's utility is known, so it leaves the expectation:

$$= \max_{C_t, w_t} \Big\{U(C_t) + \mathbb{E}_t\Big[\sum_{s=t+1}^{T} \delta^{s-t} U(C_s)\Big]\Big\}.$$

3. **Factor out δ and condition on $t+1$** ($\mathbb{E}_t[\cdot] = \mathbb{E}_t[\mathbb{E}_{t+1}[\cdot]]$):

$$= \max_{C_t, w_t} \Big\{ U(C_t) + \delta\, \mathbb{E}_t \Big[\mathbb{E}_{t+1} \Big[\sum_{s=t+1}^{T} \delta^{s-t-1} U(C_s) \Big] \Big] \Big\}.$$

4. **The inner bracket** (optimised over choices from $t+1$ on) is line 1 one date later, i.e. V_{t+1}:

$$V_t(W_t) = \max_{C_t, w_t} \big\{ U(C_t) + \delta\, \mathbb{E}_t[V_{t+1}(W_{t+1})] \big\}, \qquad V_T(W) = U(W).$$

where

$U(C_t)$	"what you eat today"; known once C_t is chosen.
$\delta\, \mathbb{E}_t[\cdot]$	"discounted, averaged over states" one period ahead.
$V_{t+1}(W_{t+1})$	"same function, one date later", evaluated at next date's wealth $W_{t+1} = (W_t - C_t)R_{p,t+1}$; random at t.
$\delta^{s-t} = \delta \cdot \delta^{s-t-1}$	how one δ is pulled outside in step 3.
$V_T(W) = U(W)$	terminal condition: "nothing follows T, so the recursion has somewhere to start"; you eat everything, $C_T = W_T$.

- "Eat something today, then land on a wealth whose future you already priced" [S9]. "The maximisation also splits: today's choice is made now, and the rest is already optimised inside V_{t+1}" [S8].
- **Traps:** forgetting δ in step 3; treating V_t as a number (it is a *function* of W, solved for every wealth level); pulling $V_{t+1}(W_{t+1})$ out of $\mathbb{E}_t$.
- **Assumptions:** time separability (lets the sum split); wealth summarises the past (returns i.i.d. or depending only on current state variables *[Beyond slides]*).

△**!** [S8] Step 3→4 moves the max over future choices *inside* $\mathbb{E}_t$. That is valid only because future choices may depend on the state that has occurred (the contingent plan of [S4]): the best plan from $t+1$ is chosen node by node. If future choices had to be fixed at t, the step would fail.

1.7 The same three steps at every date [S10]

At date	1 Choose	2 Land	3 Look up
t	eat C_t	land on W_{t+1}	read $V_{t+1}(W)$
$t+1$	eat C_{t+1}	land on W_{t+2}	read $V_{t+2}(W)$
$t+2$	eat C_{t+2}	land on W_{t+3}	read $V_{t+3}(W)$

- "One rule, written once, works at every date. Nothing about the step changes with t, only which value function it reads."
- "Tomorrow's value is what today needs, so the rows must be filled bottom-up" ⇒ backward induction.
- **Trap:** you *solve* backwards ($T \to 0$) but *live* forwards ($0 \to T$).

1.8 Backward induction [S11]

Algorithm ("solve in this order"):

1. $t = T$: $V_T(W) = U(W)$. "Nothing follows, so you simply eat everything." No optimisation.
2. $t = T-1$: $V_{T-1}(W) = \max\{U(C) + \delta\mathbb{E}[V_T(W')]\}$, using V_T.
3. …
4. $t = 1$: uses V_2.
5. $t = 0$: $V_0(W)$, "today's rule".

$$V_t(W) = \max_{C, w} \big\{ U(C) + \delta\, \mathbb{E}\, V_{t+1}(W') \big\}, \qquad t = T-1, \ldots, 0.$$

where

W'	next date's wealth $(W - C)R_p$.
V_{t+1}	already solved in the previous step of the sweep.
$t = T-1, \ldots, 0$	T one-period problems, each solved once.

- "T one-period problems, solved once each, instead of 2^T paths." At $t = 0$ you hold a rule $C_t^*(W)$, $w_t^*(W)$ for every date and every wealth level: the plan [S4] asked for.

△! [S11] The $t = T - 1$ box says "uses V_t"; it should read "uses V_T" (the $t = 1$ box correctly says "uses V_2").

Worked example *[Beyond slides]* — log utility, closed form. $U = \ln C$. Guess $V_t(W) = a_t \ln W + b_t$, with $a_T = 1$, $b_T = 0$. The one-period FOC in C:

$$\frac{1}{C} = \frac{\delta a_{t+1}}{W - C} \;\Rightarrow\; C_t^* = \frac{W_t}{1 + \delta a_{t+1}}, \qquad a_t = 1 + \delta a_{t+1} = \frac{1 - \delta^{T-t+1}}{1 - \delta}, \qquad C_t^* = W_t \, \frac{1 - \delta}{1 - \delta^{T-t+1}}.$$

where

a_t	marginal weight of $\ln W$ in V_t; how many "periods of utility" wealth still has to fund $(1 + \delta + \cdots + \delta^{T-t})$.
b_t	constant collecting expected log returns; does not affect C_t^*.
δa_{t+1}	value of a saved dollar relative to an eaten one.
C_t^*/W_t	consumption share: 1 at T; $1/(1+\delta)$ at $T-1$ (Week 6's answer); $\to 1 - \delta$ as $T \to \infty$.

- **Portfolio:** w enters only through $\delta a_{t+1} \mathbb{E}_t[\ln R_{p,t+1}]$, so the log investor maximises next period's expected log return: **myopic**, like a one-period investor. (Not true for general CRRA with predictable returns: hedging demands, Merton 1973.)
- **Numbers:** $T = 2$, $\delta = 0.96$, $W_0 = 100$: $a_2 = 1$, $a_1 = 1.96$, $a_0 = 1 + 0.96 \times 1.96 = 2.8816$, so $C_0^* = 100/2.8816 = \mathbf{34.70}$ and 65.30 is invested; at $t = 1$ eat $W_1/1.96$; at $t = 2$ eat everything.

1.9 Bellman ⇒ the Euler equation at every date [Beyond slides]

The deck goes from backward induction straight to the multi-period SDF; this is the missing link.

$$U'(C_t) = \delta \, \mathbb{E}_t\big[V'_{t+1}(W_{t+1}) \, R_{p,t+1}\big], \qquad V'_t(W_t) = U'(C_t^*).$$

where

first equation	FOC of the Bellman problem in C_t: a dollar eaten today vs the same dollar invested at $R_{p,t+1}$ and valued at V'_{t+1}.
$V'_t(W_t)$	marginal value of wealth (envelope condition): differentiate V_t in W_t holding choices fixed; it equals the right side of the FOC, hence $U'(C_t^*)$.

Shift the envelope condition one date ($V'_{t+1} = U'(C_{t+1})$) and substitute; with the weight FOC (as in W6 [S13]) it holds asset by asset:

$$U'(C_t) = \delta \, \mathbb{E}_t[U'(C_{t+1}) R_{i,t+1}] \iff \mathbb{E}_t[M_{t+1} R_{i,t+1}] = 1, \qquad M_{t+1} = \delta \frac{U'(C_{t+1})}{U'(C_t)}.$$

where

$\mathbb{E}_t$	a *conditional* expectation: it holds at every date and in every state.
M_{t+1}	Week 6's SDF with dates; known at $t+1$, random at t.
$R_{f,t} = 1/\mathbb{E}_t[M_{t+1}]$	the one-period risk-free rate, known at t but can vary over time.

- The multi-period investor prices one-period returns exactly like the Week 6 investor: "each Bellman step priced one date ahead" [S12].

1.10 The multi-period SDF: telescoping [S12]

$$M_{t+1} M_{t+2} = \delta \frac{U'(C_{t+1})}{U'(C_t)} \cdot \delta \frac{U'(C_{t+2})}{U'(C_{t+1})} = \delta^2 \frac{U'(C_{t+2})}{U'(C_t)}.$$

where

M_{t+1}, M_{t+2}	the two one-period links $t \to t+1$ and $t+1 \to t+2$.
$U'(C_{t+1})$	top of the first link, bottom of the second: "the same number, so they cancel"; date $t+1$ is "priced twice".
δ^2	one δ per period.
$U'(C_{t+2})/U'(C_t)$	only the endpoints survive.

- "Telescoping is the pricing version of backward induction: solve one link, then multiply links instead of re-solving the chain."

Why multiplying links is right *[Beyond slides]*: for a date-$(t+2)$ payoff, price backwards one period at a time:

$$P_{t+1} = \mathbb{E}_{t+1}[M_{t+2}X_{t+2}], \qquad P_t = \mathbb{E}_t[M_{t+1}P_{t+1}] = \mathbb{E}_t\big[\mathbb{E}_{t+1}[M_{t+1}M_{t+2}X_{t+2}]\big] = \mathbb{E}_t[M_{t,t+2}X_{t+2}].$$

where	
P_{t+1}	at $t+1$ the payoff is one period away: a Week 6 problem.
M_{t+1} inside $\mathbb{E}_{t+1}$	known at $t+1$, so it can move inside.
last step	tower property; same answer as pricing with the product directly.

- **Trap:** $\mathbb{E}_t[M_{t+1}M_{t+2}] \neq \mathbb{E}_t[M_{t+1}]\,\mathbb{E}_t[M_{t+2}]$ in general; it factorises when consumption growth is i.i.d. (§1.12).

1.11 The i-period kernel [S13]

$$M_{t,t+i} = M_{t+1}M_{t+2}\cdots M_{t+i} = \delta^i \frac{U'(C_{t+i})}{U'(C_t)}$$

where	
δ^i	**pure time preference**: "the further away, the less it counts, whatever happens".
$U'(C_{t+i})/U'(C_t)$	**the state adjustment**: "what a dollar is worth to you then, relative to now"; large when C_{t+i} is low.
both at once	"why it beats present value": "a distant payoff in a bad state can beat a near one in a good state".

Pricing any horizon *[Beyond slides]*:

$$P_t = \mathbb{E}_t[M_{t,t+i}X_{t+i}], \qquad B_t^{(i)} = \mathbb{E}_t[M_{t,t+i}], \qquad P_t = \mathbb{E}_t[M_{t+1}(P_{t+1}+D_{t+1})] = \sum_{i\geq 1}\mathbb{E}_t[M_{t,t+i}D_{t+i}].$$

where	
X_{t+i}	any payoff at $t+i$.
$B_t^{(i)}$	zero-coupon bond: $X = 1$.
D_{t+i}	dividend; a stock is a one-period payoff of price plus dividend.
$\sum_{i\geq 1}$	iterate forward, assuming $\mathbb{E}_t[M_{t,t+i}P_{t+i}] \to 0$ (no bubble).

Worked example *[Beyond slides]* — binomial consumption tree. $U = \ln C$, $\delta = 0.96$, $C_0 = 100$; each period consumption grows by $g_u = 1.10$ or $g_d = 0.95$ with probability $\frac{1}{2}$, independently. One-period links $M = \delta/g$: $M_u = 0.8727$, $M_d = 1.0105$, $\mathbb{E}[M] = 0.9416$, so $R_f = 1.0620$ each period. Two-period kernel $M_{0,2} = \delta^2 C_0/C_2$:

Path at $t=2$	C_2	Prob.	$M_{0,2}$	State price $q = p\,M_{0,2}$
uu	121.00	0.25	0.7617	0.1904
ud / du	104.50	0.50	0.8819	0.4410
dd	90.25	0.25	1.0212	0.2553

- Two-period zero: $B_0^{(2)} = \sum q = \mathbf{0.8867} = \mathbb{E}[M]^2$ ✓. Backward check: $B_1^{(1)} = 0.9416$ at both $t=1$ nodes, then 0.9416^2 ✓.
- "Distant bad beats near good": $M_{0,2}(\text{dd}) = 1.021 > M_{0,1}(\text{u}) = 0.873$.
- A claim to C_2 itself: $P_0 = \mathbb{E}[M_{0,2}C_2] = \delta^2 C_0 = \mathbf{92.16}$ (log utility makes $M_{0,2}C_2$ constant).
- **Traps:** $M_{t,t+i}$ depends only on the endpoints, but its *expectation* averages over all paths in between. δ^i is not the bond price: $B_t^{(i)} = \delta^i \mathbb{E}_t[U'(C_{t+i})]/U'(C_t)$.

1.12 Consequences for interest rates [Beyond slides]

$$R_{f,t} = \frac{1}{\mathbb{E}_t[M_{t+1}]}, \qquad y_t^{(i)} = \big(B_t^{(i)}\big)^{-1/i}, \qquad \text{i.i.d. links} \Rightarrow B^{(i)} = (\mathbb{E}[M])^i \Rightarrow y^{(i)} = R_f.$$

where

$R_{f,t}$	one-period gross risk-free rate, known at t.
$y_t^{(i)}$	i-period gross yield (per period).
i.i.d. links	consumption growth independent and identically distributed ⇒ flat term structure.

- In the tree above, $y^{(2)} = 0.8867^{-1/2} = 1.0620 = R_f$: flat. Non-flat curves need predictable growth or time-varying risk.
- CRRA $\gamma = 2$, same tree: $M_u = 0.96/1.1^2 = 0.7934$, $M_d = 0.96/0.95^2 = 1.0637$, $\mathbb{E}[M] = 0.9286$, $R_f = 1.0769$. More curvature with positive expected growth raises the rate.

2 Comparison tables

2.1 One period (W6) vs many periods (W7)

	Week 6	Week 7
Dates	0 and 1	$0, 1, \ldots, T$
Budget	$\tilde{C}_1 = (W_0 - C_0)\tilde{R}_p$	$W_{t+1} = (W_t - C_t)R_{p,t+1}$, every date
Objective	$U(C_0) + \delta\mathbb{E}[U(\tilde{C}_1)]$	$\mathbb{E}_0 \sum_{t=0}^{T} \delta^t U(C_t)$
Choice	two numbers (C_0, w)	a contingent plan at every node
Method	two FOCs	Bellman + backward induction
End of life	eat everything at date 1	$V_T(W) = U(W)$
SDF	$\tilde{M} = \delta U'(\tilde{C}_1)/U'(C_0)$	M_{t+1} per period; $M_{t,t+i} = \delta^i U'(C_{t+i})/U'(C_t)$
Pricing	$P = \mathbb{E}[\tilde{M}X]$	$P_t = \mathbb{E}_t[M_{t,t+i}X_{t+i}]$

2.2 Brute force vs dynamic programming [S6–S11]

	Enumerate the tree	Backward induction
Object solved	whole plan at once	one value function per date
Size	2^T terminal nodes	T one-period problems
$T = 40$	≈ 1.1 trillion nodes	40 problems
Order	any	must go $T \to 0$
Output	a plan	a rule $C_t^*(W)$, $w_t^*(W)$ for every date and wealth
Pricing analogue	re-solve every path	multiply one-period links (telescoping)

2.3 Discounting a future dollar [S13]

	Present value	SDF $M_{t,t+i}$
Depends on	date only, $1/(1+r)^i$	date (δ^i) *and* state ($U'(C_{t+i})/U'(C_t)$)
Random?	no	yes
Bad-state payoff	discounted like any other	discounted less (valued more)
Distant vs near	distant always worth less	distant bad state can beat near good state
Reduces to PV when	–	M is deterministic (no consumption risk)

2.4 The Bellman ingredients [S9]

Term	Meaning	Known at t?
$U(C_t)$	utility from eating today	yes, once chosen
$\delta\mathbb{E}_t[\cdot]$	discount one period, average over next states	operator
$V_{t+1}(W_{t+1})$	best continuation value from next date's wealth	no
$V_T(W) = U(W)$	terminal condition	–

3 Master formula sheet

Each line repeats the key symbols so the sheet can be used on its own.

Wealth [S2–S3]

- $W_{t+1} = (W_t - C_t)R_{p,t+1}$, $R_{p,t+1} = \sum_i w_{i,t}R_{i,t+1}$, $\sum_i w_{i,t} = 1$.
  where W = wealth, C = consumption, R = gross returns, w = weights.

Lifetime problem [S4]

- $\max_{C,w} \mathbb{E}_0 \sum_{t=0}^{T} \delta^t U(C_t)$ s.t. the budget, W_0 given, $C_t \geq 0$, $W_t \geq 0$.
  where δ = time preference; the choice is a contingent plan.

Value function and Bellman [S8–S11]

- $V_t(W_t) = \max \mathbb{E}_t \sum_{s=t}^{T} \delta^{s-t} U(C_s) = \max_{C_t, w_t}\{U(C_t) + \delta \mathbb{E}_t[V_{t+1}(W_{t+1})]\}$; $V_T(W) = U(W)$.
  where $\mathbb{E}_t[\mathbb{E}_{t+1}[\cdot]] = \mathbb{E}_t[\cdot]$; $\delta^{s-t} = \delta\,\delta^{s-t-1}$.
- Backward induction: solve $t = T, \ldots, 0$; T one-period problems vs 2^T paths.

Envelope and Euler *[Beyond slides]*

- $U'(C_t) = \delta \mathbb{E}_t[V'_{t+1}(W_{t+1})R_{p,t+1}]$; $V'_t(W_t) = U'(C^*_t)$.
- $\mathbb{E}_t[M_{t+1}R_{i,t+1}] = 1$, $M_{t+1} = \delta U'(C_{t+1})/U'(C_t)$, $R_{f,t} = 1/\mathbb{E}_t[M_{t+1}]$.

Multi-period SDF [S12–S13]

- $M_{t+1}M_{t+2} = \delta^2 U'(C_{t+2})/U'(C_t)$; $M_{t,t+i} = \prod_{k=1}^{i} M_{t+k} = \delta^i U'(C_{t+i})/U'(C_t)$.
  where δ^i = time preference; U' ratio = state adjustment.

Pricing *[Beyond slides]*

- $P_t = \mathbb{E}_t[M_{t,t+i}X_{t+i}]$; $B_t^{(i)} = \mathbb{E}_t[M_{t,t+i}]$; $y_t^{(i)} = (B_t^{(i)})^{-1/i}$.
- $P_t = \mathbb{E}_t[M_{t+1}(P_{t+1} + D_{t+1})] = \sum_{i\geq 1} \mathbb{E}_t[M_{t,t+i}D_{t+i}]$ (no bubble).
- i.i.d. links $\Rightarrow B^{(i)} = (\mathbb{E}[M])^i \Rightarrow$ flat yield curve.

Log utility *[Beyond slides]*

- $M_{t,t+i} = \delta^i C_t/C_{t+i}$; $V_t = a_t \ln W + b_t$, $a_t = \frac{1-\delta^{T-t+1}}{1-\delta}$; $C^*_t = W_t \frac{1-\delta}{1-\delta^{T-t+1}}$; $T \to \infty$: $C_t = (1-\delta)W_t$; portfolio myopic ($\max \mathbb{E}_t \ln R_{p,t+1}$).

4 Glossary

- **Backward induction:** solving from the last date to the first, each step using the next date's value function [S11].
- **Bellman equation:** $V_t(W) = \max\{U(C) + \delta \mathbb{E}_t V_{t+1}(W')\}$ [S9].
- **Contingent plan:** a choice of C and w at every node, fixed in advance [S4, S6].
- **Envelope condition:** $V'_t(W_t) = U'(C^*_t)$ *[Beyond slides]*.
- **Greedy strategy:** maximising each period separately; generally not optimal [S5].
- **Kernel / i-period SDF:** $M_{t,t+i} = \delta^i U'(C_{t+i})/U'(C_t)$ [S13].
- **Myopic portfolio:** chosen as if the horizon were one period (log utility) *[Beyond slides]*.
- **State variable:** the quantity summarising the past that matters for the future; here W_t.
- **Telescoping:** middle terms of a product of consecutive ratios cancel [S12].
- **Terminal condition:** $V_T(W) = U(W)$ [S9].
- **Tower property:** $\mathbb{E}_t[\mathbb{E}_{t+1}[X]] = \mathbb{E}_t[X]$ [S8].
- **Value function:** $V_t(W)$, best expected discounted utility from t given wealth W [S8].
- **Wealth evolution:** $W_{t+1} = (W_t - C_t)R_{p,t+1}$ [S3].
- **Zero-coupon bond price:** $B_t^{(i)} = \mathbb{E}_t[M_{t,t+i}]$ *[Beyond slides]*.

5 Self-test

Questions

Q1 ([S4, S6]). Why must the solution specify C and w at every node, not just along the realised path? What does "one rule for every state, chosen once" mean?

Q2 ([S7, S11]). With two outcomes per date, how many nodes are at date $T = 15$? How many one-period problems does backward induction solve? Why can't it start at $t = 0$?

Q3 ([S8–S9]). Derive the Bellman equation from the definition of V_t. Name the property used in each step and state the terminal condition.

Q4 ([S2–S3]). $W_t = 100$, $C_t = 30$; 60% in a stock with $R = 1.20$ or 0.90, 40% in a bond with $R_f = 1.05$. Compute W_{t+1} in both states.

Q5 ([S11]). Log utility, $T = 2$, $\delta = 0.95$. (a) Show $C^*_{T-1} = W_{T-1}/(1+\delta)$. (b) Show $a_t = 1 + \delta a_{t+1}$ and find C^*_0 for $W_0 = 200$. (c) Find C_t/W_t as $T \to \infty$.

Q6 ([S12–S13]). Show $M_{t+1}M_{t+2}M_{t+3} = \delta^3 U'(C_{t+3})/U'(C_t)$, and that pricing a date-$(t+2)$ payoff backwards gives $P_t = \mathbb{E}_t[M_{t,t+2}X_{t+2}]$.

Q7 ([S13]). Log utility, $\delta = 0.96$, $C_0 = 100$, growth 1.10 or 0.95 with probability $\frac{1}{2}$ each period (i.i.d.). (a) M_u, M_d, R_f. (b) $M_{0,2}$ on each path and $B_0^{(2)}$. (c) Verify $B_0^{(2)} = (\mathbb{E}[M])^2$ and find the two-period yield.

Q8 (traps). True or false? (a) "δ^i is the price of a bond paying \$1 in i periods." (b) "A further-away payoff is always worth less." (c) "V_t is a number computed once." (d) "Maximising $U(C_t)$ every period maximises lifetime utility." (e) "$\mathbb{E}_t[M_{t+1}M_{t+2}] = \mathbb{E}_t[M_{t+1}]\mathbb{E}_t[M_{t+2}]$ always."

Q9 (*[Beyond slides]*). Use the FOC in C_t and the envelope condition to show $\mathbb{E}_t[M_{t+1}R_{i,t+1}] = 1$ at every date.

Q10 ([S12]). Explain "telescoping is the pricing version of backward induction". Repeat Q7(a) with CRRA $\gamma = 2$ and explain why R_f changes.

Solutions

S1. You don't know at date 0 which branch occurs, so the plan must say what to do at each one; each node's choice affects expected utility through its probability. "Chosen once" means the optimiser picks a whole rule (a function of the state) at $t = 0$ and later just reads it off.

S2. $2^{15} = \mathbf{32{,}768}$ nodes. **15** one-period problems (dates $14, \ldots, 0$; date 15 is $V_{15} = U$). At $t = 0$ you need V_1, which needs V_2, …; only $V_T = U$ is known without solving.

S3. (1) Definition; (2) $U(C_t)$ leaves $\mathbb{E}_t$ because C_t is chosen at t; (3) $\delta^{s-t} = \delta\,\delta^{s-t-1}$ and tower property $\mathbb{E}_t = \mathbb{E}_t\mathbb{E}_{t+1}$; (4) the inner bracket, maximised over state-contingent choices from $t+1$, is $V_{t+1}(W_{t+1})$. Terminal: $V_T(W) = U(W)$.

S4. Invested 70. $R_p = 0.6R_s + 0.42$: up 1.14, down 0.96. $W_{t+1} = \mathbf{79.8}$ or $\mathbf{67.2}$.

S5. (a) $V_T = \ln W$; $1/C = \delta/(W-C) \Rightarrow C = W/(1+\delta)$. (b) $1/C = \delta a_{t+1}/(W-C) \Rightarrow W - C = \delta a_{t+1}C$, and $V_t = (1+\delta a_{t+1})\ln W + \text{const}$, so $a_t = 1 + \delta a_{t+1}$. $a_1 = 1.95$, $a_0 = 2.8525$, $C^*_0 = 200/2.8525 = \mathbf{70.11}$. (c) $a_t \to 1/(1-\delta)$, so $C_t/W_t \to 1 - \delta = \mathbf{5}\%$.

S6. The $U'(C_{t+1})$ and $U'(C_{t+2})$ cancel, leaving $\delta^3 U'(C_{t+3})/U'(C_t)$. With $P_{t+1} = \mathbb{E}_{t+1}[M_{t+2}X_{t+2}]$:

$$P_t = \mathbb{E}_t[M_{t+1}P_{t+1}] = \mathbb{E}_t[\mathbb{E}_{t+1}[M_{t+1}M_{t+2}X_{t+2}]] = \mathbb{E}_t[M_{t,t+2}X_{t+2}].$$

S7. (a) $M_u = \mathbf{0.8727}$, $M_d = \mathbf{1.0105}$, $\mathbb{E}[M] = 0.9416$, $R_f = \mathbf{1.0620}$. (b) uu 0.7617, ud/du 0.8819, dd 1.0212; $B_0^{(2)} = \mathbf{0.8867}$. (c) $0.9416^2 = 0.8867$ ✓; $y^{(2)} = 0.8867^{-1/2} = \mathbf{1.0620}$ (flat).

S8. (a) False: $B_t^{(i)} = \delta^i \mathbb{E}_t[U'(C_{t+i})]/U'(C_t)$. (b) False: a distant bad-state payoff can beat a near good-state one ($1.021 > 0.873$ in Q7). (c) False: a function of W. (d) False: greedy; must include $\delta\mathbb{E}_t[V_{t+1}]$. (e) False: $\mathbb{E}_t[M_{t+1}\mathbb{E}_{t+1}[M_{t+2}]]$; factorises only if $\mathbb{E}_{t+1}[M_{t+2}]$ is non-random.

S9. FOC: $U'(C_t) = \delta\mathbb{E}_t[V'_{t+1}R_{p,t+1}]$. Envelope: $V'_t(W_t) = \delta\mathbb{E}_t[V'_{t+1}R_{p,t+1}] = U'(C_t)$, so $V'_{t+1} = U'(C_{t+1})$. Substitute: $U'(C_t) = \delta\mathbb{E}_t[U'(C_{t+1})R_{p,t+1}]$; the weight FOC makes it hold per asset; divide by $U'(C_t)$. ■

S10. Backward induction solves the last link first and reuses each value function instead of re-solving the future; telescoping prices each one-period link once and multiplies them, so any horizon is priced by chaining one-period steps. CRRA: $M_u = 0.7934$, $M_d = 1.0637$, $\mathbb{E}[M] = 0.9286$, $R_f = \mathbf{1.0769}$ (up from 1.0620): with

expected growth, a more curved U strengthens the wish to borrow against the richer future, so a higher rate is needed.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title: Multi-Period Asset Pricing	header
S2	One period: $W_{t+1} = (W_t - C_t)R_{p,t+1}$	1.1
S3	Many periods; $R_{p,t+1} = \sum_i w_{i,t} R_{i,t+1}$	1.2
S4	Lifetime problem; constraints; "one rule for every state"	1.3
S5	"The best grade every term is not the best life"	1.4
S6	Possible paths: 2^t nodes; plan at every node	1.5
S7	Work explodes with the horizon △!	1.5
S8	Value function: four-line Bellman derivation △!	1.6
S9	Bellman equation; components; $V_T(W) = U(W)$	1.6
S10	Choose → land → look up; fill bottom-up	1.7
S11	Backward induction △!	1.8
S12	Two-period SDF; telescoping	1.10
S13	i-period kernel; time preference × state adjustment	1.11

Appendix B. Flag summary (△!)

1. **Notation:** W_t is wealth here (and in W6), not a Brownian motion as in QF620.
2. **[S7]** Chart not to scale; each +10 in T multiplies nodes by ≈ 1000. The $T = 10$ point (1,024) is unlabelled.
3. **[S8]** Step 3→4 moves the max inside $\mathbb{E}_t$; valid only because future choices are state-contingent.
4. **[S11]** The $t = T - 1$ box says "uses V_t"; should be "uses V_T".
5. **[S11]–[S12]** The Euler equation is never derived from the Bellman equation; the envelope condition $V_t'(W) = U'(C^*)$ (§1.9) is the missing step.
6. **[S12]–[S13]** M_{t+1} is used without being redefined (W6's $\tilde{M}$ with dates); pricing uses *conditional* expectations $\mathbb{E}_t$.
7. **[S4], [S9]** δ^t discounts utility, not dollars; δ^i is not a bond price.
8. **[S4]** Finite horizon, no bequest motive; $V_T(W) = U(W)$ depends on that.